

Word-Level Language Model for *Poirot Investigates*

This project implements a word-level neural language model trained on *Poirot Investigates* by Agatha Christie. The objective was to build a recurrent neural network capable of predicting the next word in a sequence and generating stylistically similar text when prompted.

Data Processing

The raw text was first cleaned by removing Project Gutenberg metadata. Unicode characters (curly quotes, em dashes, non-breaking spaces) were normalised to ensure consistent tokenisation. The text was tokenised at the word level using a regex based tokenizer that preserved punctuation as separate tokens. An explicit <eos> token was inserted after sentence ending punctuation to help the model learn sentence boundaries.

A vocabulary was constructed using a minimum frequency threshold (MIN_FREQ = 1) and capped at 12,000 tokens to balance coverage and model stability. Special tokens (<pad>, <unk>, <eos>) were included. The words (~53k words) was split into 90% training and 10% validation data. Sliding windows of length 50 (SEQ_LEN = 50) were used to create input-target pairs for next word prediction.

Model Architecture and Training

A word-level LSTM language model was implemented in TensorFlow. The architecture consists of:

- Embedding layer (128 dimensions)
- Single LSTM layer (256 units)
- Layer Normalisation
- Dropout (0.4)
- Dense output layer projecting to vocabulary logits

The model was trained using sparse categorical cross-entropy loss with the Adam optimiser (LR = $3e-4$) and gradient clipping. Early stopping and learning rate reduction on plateau were applied to prevent overfitting. The best performing checkpoint (based on validation loss) was saved for inference.

Increasing the context window from 30 to 50 tokens improved narrative consistency. Adding <eos> tokens improved sentence termination behaviour and controlled nucleus (top-p) sampling reduced punctuation loops during generation.

Text Generation Quality

The model successfully generates multi-sentence outputs containing stylistically consistent vocabulary (example: "Poirot," "Hastings," "Prime Minister"). It captures local structure and narrative rhythm but exhibits grammatical inconsistencies and occasional semantic drift. These limitations are expected due to:

- Small training data (~53k words)
- Word level tokenisation
- Single book domain training
- LSTM architecture without large-scale pretraining

Comparison with Large Language Models

When compared to a large pretrained model like ChatGPT, the latter produces grammatically good, semantically coherent and globally structured text. This difference is due to transformer architecture, subword tokenisation and large scale pretraining on diverse corpus. In contrast, the implemented model learns only local stylistic patterns from a limited dataset without external knowledge.

Overall, the model demonstrates successful next word prediction and stylistic text generation within the constraints of dataset size and architecture design.

Prompt	LSTM Model Output	ChatGPT Output	Key Differences
Poirot looked at me and said	"Poirot looked at me and said, and then in the own next - room, and the letter he should go back to the lift."	"Poirot looked at me and said, 'My dear Hastings, you are in far too great a hurry. The truth does not reveal itself to those who rush past it.' He adjusted his cuffs with quiet precision and regarded me with a knowing smile."	The LSTM captures structural elements (dialogue opening, narrative tone) but contains grammatical errors ("in the own next - room") and unclear meaning. ChatGPT maintains grammatical accuracy, logical progression and consistent character voice.
Hastings, you must understand	"Hastings, you must understand that the world is quite always said. I should asked myself in the road, and I know you that you have have gone to find my word."	"Hastings, you must understand that the world rarely presents its mysteries plainly. We must observe carefully, question patiently, and trust that the smallest detail may carry the greatest significance."	The LSTM demonstrates thematic similarity (reflection, explanation tone) but struggles with verb agreement ("should asked," "have have") and semantic clarity. ChatGPT produces coherent reasoning, controlled syntax and sustained narrative voice.